

Can Large Language Models Assess Bipolar Disorder Mood States? A Validation Study Against Clinically Annotated Social Media Data

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Abstract. Bipolar disorder (BD) is frequently misdiagnosed as major depressive disorder (MDD), and accurate mood state monitoring is critical for timely intervention. Large language models (LLMs) have shown promise in clinical text analysis, yet their ability to replicate expert-level psychiatric mood assessment from social media remains underexplored. We evaluate whether an LLM (Gemini 2.5 Pro) can classify mood states and estimate mood intensity from Reddit posts, using the clinically validated BD-Risk dataset (274 expert-annotated posts) as a benchmark. Through a DSM-5-aligned prompting framework, the LLM classifies each post into four mood states and assigns a 7-point mood intensity score. Results reveal a stark asymmetry: the LLM achieves high recall for depressive states (94.3%) but critically underdetects hypomania (26.1%) and mania (18.8%), with a systematic negative bias (-0.573 on the mood scale). Mood intensity ratings show moderate agreement with experts (Spearman $\rho = 0.60$, quadratic weighted $\kappa = 0.60$), with strong performance on the depressive pole (± 1 accuracy: 90.9%) but substantial degradation on the manic pole (64.6%). Error analysis identifies six clinically significant failure patterns, including the masking of suicidal ideation under calm writing styles and conflation of retrospective behavioral descriptions with current emotional tone. These findings highlight both the potential and the boundaries of LLM-based mood assessment, providing actionable insights for prompt engineering and clinical NLP system design.

Keywords: Bipolar Disorder · Large Language Models · Mood Assessment · Social Media · Clinical NLP · Mental Health

1 Introduction

Bipolar disorder (BD) is a chronic psychiatric condition characterized by recurrent episodes of mania, hypomania, and depression, affecting approximately 1 to 2% of the global population [1]. The disorder carries a substantially elevated suicide risk and imposes significant functional impairment across social, occupational, and cognitive domains [2]. A critical challenge in BD management is diagnostic delay: an estimated 17 to 50% of BD cases are initially misdiagnosed as major depressive disorder (MDD), primarily because patients tend to seek help during depressive episodes and may not recognize manic or hypomanic states as pathological

[3]. This misdiagnosis can lead to inappropriate treatment (e.g., antidepressant monotherapy, which may trigger manic switching) and prolonged suffering.

The emergence of social media as a venue for mental health disclosure has created new opportunities for computational approaches to mood monitoring. Platforms such as Reddit host communities (e.g., r/bipolar, r/BipolarReddit) where individuals with BD openly discuss symptoms, treatment experiences, and daily functioning. Prior work has explored automated detection of mental health conditions from social media text, including BD and MDD classification [4–6], but these approaches typically focus on binary diagnosis prediction rather than fine-grained mood state assessment.

Large language models (LLMs) have demonstrated impressive capabilities in clinical text understanding and annotation tasks [7, 8]. Their ability to process nuanced language, follow complex clinical criteria, and produce structured outputs makes them attractive candidates for automating psychiatric assessment tasks that traditionally require expert human judgment. However, the extent to which LLMs can reliably replicate expert-level mood state classification, particularly for the manic pole of BD, which is notoriously difficult to detect from text, remains an open question with direct clinical safety implications.

In this work, we systematically evaluate the ability of a state-of-the-art LLM (Gemini 2.5 Pro) to assess BD mood states from individual Reddit posts, using the BD-Risk dataset [9] as an external benchmark. This dataset provides mood labels assigned by two clinical experts (a psychiatrist and a clinical psychologist) on a 7-point scale (−3 to +3), making it uniquely suited for validating automated mood assessment at clinical granularity.

Our contributions are threefold:

1. We design a DSM-5-aligned prompting framework for post-level mood state classification and intensity estimation, producing structured outputs across four mood states and a 7-point scale.
2. We conduct a comprehensive validation against 274 expert-annotated posts, reporting agreement metrics at two granularities (categorical state and ordinal intensity) with subgroup analyses by diagnostic group (MDD vs. MDD→BD).
3. We present a detailed error taxonomy identifying six clinically significant failure patterns, including safety-critical blind spots in suicidal ideation detection.

2 Related Work

2.1 Social Media in Mental Health Detection

The use of social media data for mental health research has grown substantially over the past decade. Pioneering work by De Choudhury et al. [10] demonstrated that linguistic and behavioral signals on social media can predict the onset of depression. Subsequent efforts have expanded to multiple conditions: the SMHD dataset [4] covers nine mental health conditions using self-reported diagnoses,

while Coppersmith et al. [5] established shared tasks for depression and PTSD detection from Twitter.

For bipolar disorder specifically, Sekuli'c et al. [6] proposed Reddit-based BD classification, and Jagfeld et al. [11] compiled a large-scale BD Reddit corpus. However, these datasets rely on self-reported diagnoses without clinical expert validation, limiting their reliability [12, 13]. The BD-Risk dataset [9] addresses this gap by providing mood labels validated by a psychiatrist and a clinical psychologist, making it a valuable benchmark for our evaluation.

2.2 LLMs for Clinical NLP and Mental Health

Large language models have been increasingly applied to mental health NLP tasks. Xu et al. [7] evaluated multiple LLMs on mental health prediction from online text, finding that while LLMs show promise, their performance varies substantially across conditions and datasets. Yang et al. [8] developed MentalLLaMA, a model fine-tuned for interpretable mental health analysis on social media. In the BD-Risk paper itself, Lee et al. [9] reported that ChatGPT achieved only 0.130 F1 on BD risk detection, far below their proposed multi-task learning model (0.578 F1), highlighting the difficulty of this task for general-purpose LLMs.

Our work differs from prior LLM evaluations in two key aspects. First, rather than predicting binary diagnosis, we evaluate fine-grained mood state and intensity estimation, a more clinically informative task. Second, we focus specifically on identifying *where* and *how* the LLM fails, producing an actionable error taxonomy rather than aggregate performance metrics alone.

3 Methodology

3.1 The BD-Risk Dataset

We use the BD-Risk dataset introduced by Lee et al. [9] at NAACL 2024 as our evaluation benchmark. The dataset comprises 7,346 Reddit posts from 1,025 users collected from mental health subreddits (r/Depression, r/bipolar, r/BipolarReddit, r/BipolarSOs) spanning 2008 to 2023. Users are divided into two clinically defined groups: MDD-only (569 users, diagnosed with MDD without subsequent BD diagnosis) and MDD→BD (456 users, initially diagnosed with MDD and later rediagnosed with BD).

Each post carries a mood level label on a 7-point scale (-3 to $+3$), where negative values indicate depressive mood and positive values indicate manic mood. Labels were assigned by trained annotators under the guidance of a psychiatrist and subsequently validated by two clinical experts (a psychiatrist E1 and a clinical psychologist E2). The expert validation on 150 randomly sampled posts yielded a Krippendorff's α of 0.87 and inter-expert Cohen's κ values of 0.63 to 0.69 [9], establishing the reliability of the gold labels.

The full dataset exhibits a heavily skewed mood distribution: 88.9% of posts carry mood levels ≤ 0 (depressive to stable range), while only 6.2% have mood

levels > 0 (hypomanic to manic range). To ensure adequate representation of both mood poles in this pilot evaluation, we employed stratified sampling: approximately half of the sample was drawn from posts with mood levels ≤ 0 , and the other half from posts with mood levels > 0 . This oversampling of the minority manic pole enables meaningful evaluation across the full mood spectrum. After LLM processing, our validation set consists of **274 posts** (110 with mood ≤ 0 , 164 with mood > 0 ; 116 from the MDD group, 158 from the MDD $\rightarrow$ BD group).

3.2 DSM-5 Aligned Labeling Framework

We develop a structured prompting framework that aligns LLM mood assessment with DSM-5 clinical criteria [14]. The framework requires the LLM to produce two levels of output for each post:

State Classification (4-class). The LLM assigns each post to one of four mood states (Depressive, Stable, Hypomanic, or Manic) or marks it as Uncertain when the post lacks sufficient information for assessment. Each state is defined through both clinical markers (drawn from DSM-5 episode criteria) and their observable linguistic correlates:

- **Manic:** Grandiosity, pressured writing (run-on sentences, excessive capitalization), flight of ideas (tangential topic shifts), extreme irritability or euphoria.
- **Hypomanic:** Elevated energy and pace with maintained coherence, social disinhibition, uncharacteristic intensity without psychotic features.
- **Depressive:** Linguistic constriction, absolutist language (“never,” “nothing”), high self-focus (first-person pronouns), cognitive distortions, suicidal ideation.
- **Stable:** Balanced emotional tone, metacognitive reflection, proportionate responses, community-supportive language.

Mood Intensity (7-point scale). The LLM assigns an integer score from -3 (severe depressive crisis) to $+3$ (severe manic expression), with explicit boundary criteria for adjacent levels. The prompt provides anchor definitions: -3 requires explicit suicidality or psychological collapse; -2 indicates marked functional impairment; -1 reflects mild depressive tone; 0 represents stable baseline; $+1$ indicates mild positive activation; $+2$ requires clear manic-side activation; and $+3$ is reserved for extreme manic intensity with psychotic features or severe disorganization.

The framework also supports a `with_mixed_features` specifier (following DSM-5 mixed-features criteria), but this element is not present in the BD-Risk gold labels and is therefore excluded from the quantitative evaluation.

Mapping between scales. The BD-Risk dataset provides only the 7-point mood label; categorical states are not directly annotated. To enable state-level comparison, we derive gold states from mood labels using the mapping defined in the BD-Risk paper: mood labels -3 , -2 , and -1 map to Depressive; 0 maps to Stable; $+1$ maps to Hypomanic; and $+2$ and $+3$ map to Manic. This derived mapping introduces a source of granularity mismatch, as discussed in Section 4.

3.3 LLM Configuration

We use Gemini 2.5 Pro [15] as our LLM, accessed through the official API with structured JSON output. Each post is processed independently (zero-shot, no few-shot examples), with the full DSM-5-aligned prompt provided as a system instruction. The model is configured to return a JSON object containing **state**, **mood_label**, **specifiers**, **confidence** (High/Medium/Low), and **reasoning** fields. Temperature is set to the default value. No fine-tuning or in-context learning from BD-Risk labels is performed.

3.4 Evaluation Metrics

Throughout this paper, we refer to the clinician-assigned labels from the BD-Risk dataset as gold labels and the LLM outputs as predictions.

We evaluate at two granularities:

State classification: Accuracy, per-class precision/recall/F1, macro F1, Cohen’s κ [16] (unweighted and linearly weighted).

Mood intensity: Exact match accuracy, ± 1 tolerance accuracy, mean absolute error (MAE), root mean squared error (RMSE), mean bias (prediction – gold), Spearman rank correlation (ρ), Pearson correlation (r), and quadratic weighted Cohen’s κ .

All metrics are computed both overall and stratified by diagnostic subgroup (MDD vs. MDD→BD).

4 Results

4.1 State Classification

Table 1 presents the per-class classification metrics for the four-state categorization. The results reveal a pronounced asymmetry between the depressive and manic poles.

Table 1. Per-class state classification metrics (excluding 20 Uncertain outputs). Overall accuracy: 46.7% (incl. Uncertain) / 50.4% (excl. Uncertain). Cohen’s $\kappa = 0.314$.

State	Precision	Recall	F1	Support
Depressive	0.582	0.943	0.719	87
Stable	0.127	0.563	0.207	16
Hypomanic	0.861	0.261	0.400	119
Manic	1.000	0.188	0.316	32
Macro avg.	0.642	0.489	0.410	254
Weighted avg.	0.673	0.504	0.487	254

The LLM demonstrates high recall for Depressive states (94.3%) but at the cost of precision (58.2%), indicating systematic over-prediction of depression. Conversely,

Hypomanic and Manic states show high precision (86.1% and 100%, respectively) but critically low recall (26.1% and 18.8%), meaning the LLM correctly identifies these states when it does predict them, but misses the vast majority of cases. The overall Cohen’s κ of 0.314 indicates fair agreement.

Table 2 presents the confusion matrix including the Uncertain category. Two dominant error patterns are visible: (1) 41 of 128 gold-label Hypomanic posts are predicted as Depressive, and another 47 as Stable; (2) 11 of 36 gold-label Manic posts are predicted as Depressive.

Table 2. State confusion matrix (rows: gold label; columns: prediction). DEP = Depressive, STA = Stable, HYP = Hypomanic, MAN = Manic, UNC = Uncertain. Bold values indicate correct classifications.

Gold \ Pred.	DEP	STA	HYP	MAN	UNC	Total
Depressive	82	5	0	0	1	88
Stable	7	9	0	0	6	22
Hypomanic	41	47	31	0	9	128
Manic	11	10	5	6	4	36

4.2 Mood Intensity Assessment

Table 3 presents the overall mood intensity metrics. While exact match accuracy is modest (40.2%), the ± 1 tolerance accuracy reaches 74.8%, and the Spearman correlation of 0.60 ($p < 0.001$) indicates moderate-to-strong ordinal agreement. The quadratic weighted κ of 0.604 represents substantial agreement on the ordinal scale, notably higher than the nominal κ of 0.314 for state classification.

Table 3. Overall mood intensity assessment metrics ($n = 274$).

Metric	Value
Exact match accuracy	40.2% (110/274)
± 1 tolerance accuracy	74.8% (205/274)
Mean absolute error (MAE)	1.026
Root mean squared error (RMSE)	1.539
Mean bias (Pred. – Gold)	−0.573
Spearman ρ	0.603 ($p < 0.001$)
Pearson r	0.640 ($p < 0.001$)
Quadratic weighted κ	0.604

The systematic negative bias of -0.573 confirms that the LLM consistently underestimates mood toward the depressive pole. This bias is not uniform across the scale: Table 4 shows a dramatic performance gap between the depressive and manic poles.

Table 4. ± 1 tolerance and exact match accuracy by mood polarity.

Mood range	n	± 1 Acc.	Exact Acc.
Depressive (-3 to -1)	88	90.9%	58.0%
Neutral (0)	22	86.4%	59.1%
Manic ($+1$ to $+3$)	164	64.6%	28.0%

For depressive posts (gold labels -3 to -1), the ± 1 accuracy is 90.9%, comparable to the inter-expert agreement reported in the BD-Risk study. For manic-pole posts (gold labels $+1$ to $+3$), the ± 1 accuracy drops to 64.6%, with exact match accuracy of only 28.0%. The per-label breakdown (Table 5) shows that performance degrades monotonically with increasing positive mood intensity: from 73.4% ± 1 accuracy at $+1$ to 20.0% at $+3$.

Table 5. Mood intensity accuracy by gold label.

Gold label	n	Exact	± 1	± 1 Acc.
-3	34	25	31	91.2%
-2	46	24	42	91.3%
-1	8	2	7	87.5%
0	22	13	19	86.4%
$+1$	128	37	94	73.4%
$+2$	26	8	10	38.5%
$+3$	10	1	2	20.0%

4.3 Subgroup Analysis

Table 6 compares performance between the MDD and MDD $\rightarrow$ BD diagnostic groups. The MDD $\rightarrow$ BD group shows higher state classification accuracy (55.2% vs. 44.1%) and higher mood exact match accuracy (44.3% vs. 34.5%). However, the MDD $\rightarrow$ BD group also exhibits a stronger negative bias (-0.690 vs. -0.414), suggesting that the LLM more severely underestimates manic-pole mood in this group, which contains more manic-range posts by definition.

Table 6. Performance by diagnostic subgroup.

Group	n	State Acc.	Mood ± 1	Mood Bias
MDD	116	44.1%	75.0%	-0.414
MDD $\rightarrow$ BD	158	55.2%	74.7%	-0.690

4.4 The Uncertain Label as Quality Control

The LLM produced Uncertain outputs for 20 posts (7.3%), exercising its option to abstain when post content was deemed insufficient for mood assessment. Analysis of these posts reveals that 19 of 20 carry non-depressive gold labels (Stable: 6, Hypomanic: 9, Manic: 4), with only 1 depressive case. The Uncertain posts share common characteristics: pharmaceutical inquiries, literary quotations shared without personal commentary, and extremely brief or fragmented text.

This abstention mechanism functions as a quality control filter, preventing forced labeling when evidence is insufficient. However, one notable exception (a post containing the phrase “I want to die” embedded within metacommentary about depression) reveals that the Uncertain label can occasionally mask clinically significant signals, a pattern discussed further in Section 5.

5 Error Analysis

We identify six systematic error patterns through analysis of 69 mood outliers (posts where $|\text{Prediction} - \text{Gold}| > 1$, comprising 25.2% of the dataset). Of the 20 most extreme outliers, 16 involve manic or hypomanic posts misclassified as depressive, confirming the systematic manic-pole underdetection observed in the aggregate metrics.

5.1 Pattern 1: Retrospective Behavioral Cues Ignored

The dominant error pattern (43 of 69 outliers). When users describe manic-episode behaviors (impulsive spending, aggressive confrontations, hyperactivity) in a retrospective post written with remorse or self-blame, the LLM assigns depressive labels based on the *current emotional tone* rather than recognizing the *clinical significance of the described behaviors*.

Representative case: A user describes uncontrollable impulsive spending that depleted their savings, expressing intense self-blame and calling themselves “worthless.” The gold label is +2 (Manic), recognizing the spending as a hallmark manic symptom. The prediction is −2 (Depressive), focusing on the self-deprecating language. This pattern reflects a fundamental limitation: the LLM processes text as sentiment rather than as clinical evidence, conflating the *affect of the narrator* with the *clinical state being described*.

5.2 Pattern 2: Mixed Features Conflated with Neutrality

When posts exhibit symptoms from both poles simultaneously (e.g., severe sleep disruption and inability to concentrate alongside aggressive outbursts), the LLM sometimes assigns a neutral score (mood label 0), treating the coexistence of opposing signals as cancellation rather than as a clinically significant mixed episode. Under DSM-5 criteria [14], mixed features should be annotated as a specifier on the primary state without altering the severity rating of that state.

5.3 Pattern 3: Calm Writing Style Masking Suicidal Ideation

A safety-critical failure pattern. Certain posts express suicidal ideation (“I want to die”) within a calm, reflective, or educational writing style. The LLM, treating the overall linguistic register as indicative of emotional stability, classifies these posts as Stable (mood label 0). The gold label is -3 (severe depressive crisis), correctly prioritizing the suicidal content over the writing style. This finding has direct implications for clinical deployment: *literary sophistication and emotional detachment in writing do not rule out severe psychiatric symptoms*, and LLM-based systems must be explicitly engineered to detect crisis-level signals regardless of surrounding discourse register.

5.4 Pattern 4: End-of-Post Hope Overriding Pervasive Impairment

Posts describing severe functional impairment (academic collapse, inability to maintain daily routines, social withdrawal) sometimes conclude with a single hopeful statement (e.g., “my therapist said maybe I shouldn’t give up”). The LLM anchors on this terminal positive signal and classifies the post as Stable, while the gold label reflects the pervasive impairment evidence throughout the post. This suggests a recency bias in the LLM’s text processing.

5.5 Pattern 5: Substance-Induced Activation vs. Endogenous Mood

When users describe mood elevation caused by substances (e.g., caffeine-induced euphoria described as feeling “on top of the world”), the LLM may interpret these descriptions as evidence of hypomanic mood. The gold labels distinguish between pharmacologically induced activation and endogenous mood states, reflecting the user’s baseline mood rather than transient substance effects. This boundary case is not explicitly addressed in current DSM-5-based prompts.

5.6 Pattern 6: Uncertain Masking Embedded Clinical Signals

In rare cases (1 of 20 Uncertain posts), the LLM correctly identifies a post as metacommentary or informational but fails to recognize clinically significant content embedded within it. A post critiquing the “aestheticization of depression” online contained explicit suicidal statements that were masked by the Uncertain classification. While Uncertain functions appropriately in 95% of cases, this pattern highlights the need for a safety-override mechanism that detects crisis-level language regardless of overall post classification.

6 Discussion

6.1 Asymmetric Performance and Its Clinical Implications

Our results demonstrate a clear dichotomy in LLM mood assessment capability. On the depressive pole, the LLM achieves performance approaching expert-level

agreement: ± 1 accuracy of 90.9% for depressive posts and Depressive recall of 94.3%. On the manic pole, performance drops dramatically, with ± 1 accuracy of 64.6% and Manic recall of 18.8%.

This asymmetry likely reflects training data distribution: depressive language is more prevalent in general text corpora and has more stereotypical linguistic markers (negativity, self-focus, hopelessness). Manic and hypomanic states, by contrast, manifest through behavioral patterns (impulsive actions, decreased sleep need, grandiose plans) that may be described in language carrying *any* emotional valence, including regret, confusion, or humor, making them harder to detect through surface-level linguistic analysis.

6.2 Comparison with Human Agreement and Prior Models

The BD-Risk dataset reports inter-expert Cohen’s κ values of 0.63 to 0.69 for mood level annotation [9]. Our LLM achieves a quadratic weighted κ of 0.604 for mood intensity, approaching the lower bound of expert-expert agreement. However, this aggregate metric masks the polarity asymmetry: agreement on the depressive pole likely exceeds expert levels, while agreement on the manic pole falls substantially below.

The LLM’s Cohen’s κ of 0.314 for state classification (4-class) is notably lower, reflecting both the genuine difficulty of the task and the fact that gold states are *derived* from mood labels rather than directly annotated. The mapping from 7-point mood labels to 4-class states introduces boundary effects (e.g., a mood label of +1 maps to Hypomanic, but this boundary is clinically ambiguous).

For mood level prediction, Lee et al. [9] reported an MAE of 0.701 for their multi-task learning (MTL) model, compared to our LLM’s MAE of 1.026. However, a direct comparison requires several caveats. Their MTL model operates at the *user level*, leveraging longitudinal sequences of posts per user and jointly learning BD risk prediction, mood level estimation, and mood variation assessment through supervised training on the full dataset. In contrast, our approach performs *post-level* zero-shot inference without any training on BD-Risk data, access to user history, or task-specific fine-tuning. Despite this more constrained setting, the LLM achieves moderate ordinal agreement ($\kappa_w = 0.604$, Spearman $\rho = 0.60$), suggesting that LLMs can extract meaningful mood signals from individual posts even without longitudinal context or supervised adaptation.

6.3 Implications for Prompt Engineering

Our error analysis suggests several concrete improvements for DSM-5-aligned prompts:

1. **Behavioral evidence weighting:** Explicitly instruct the LLM to evaluate described behaviors (spending sprees, aggressive outbursts, hyperactivity) as clinical evidence independent of the narrator’s current emotional tone.
2. **Suicidal ideation override:** Implement a safety-priority rule: the presence of suicidal language should always trigger depressive classification regardless of surrounding discourse register.

3. **Mixed-features calibration:** Clarify that mixed-features specifiers should not influence mood intensity scores toward neutrality.
4. **Substance-mood disambiguation:** Add explicit guidance for distinguishing pharmacologically induced activation from endogenous mood elevation.

6.4 Limitations

Several limitations should be noted. First, we evaluate a single LLM (Gemini 2.5 Pro); the error patterns identified may be model-specific. Second, our evaluation is at the post level, whereas clinical mood assessment typically considers longitudinal patterns. Third, gold states are derived from mood labels through a predefined mapping, introducing potential misalignment with how experts conceptualize categorical states. Fourth, our evaluation uses a stratified subset of 274 posts (from 7,346 total) with deliberate oversampling of the manic pole; performance on a representative sample may differ, particularly for the depressive pole where only 110 posts were included despite comprising 88.9% of the full dataset. The extreme mood levels also remain underrepresented (+3: $n = 10$; -1: $n = 8$). Finally, as with all social media based research, the ecological validity of Reddit posts as proxies for clinical mood states requires further investigation.

7 Conclusion

We have presented a systematic evaluation of LLM-based mood state assessment for bipolar disorder using a clinically validated benchmark. Our findings reveal that while LLMs demonstrate strong capability in detecting depressive states from social media text, they exhibit a critical blind spot for manic and hypomanic states, precisely the states that distinguish BD from MDD and that are most important for timely clinical intervention.

The six error patterns identified in this study provide a concrete roadmap for improving LLM-based clinical assessment systems. Most critically, the finding that calm or reflective writing styles can mask suicidal ideation from LLM detection underscores the need for robust safety mechanisms in any clinical deployment of these systems.

Future work should explore multi-model comparison to determine whether manic-pole underdetection is a general LLM limitation or model-specific, prompt optimization guided by the error taxonomy presented here, and extension to longitudinal (multi-post) mood trajectory assessment, which may provide richer clinical signals than single-post analysis.

References

1. Grande, I., Berk, M., Birmaher, B., Vieta, E.: Bipolar Disorder. *The Lancet*. 387, 1561–1572 (2016).

2. Vieta, E., Berk, M., Schulze, T.G., Carvalho, A.F., Suppes, T., Calabrese, J.R., Gao, K., Miskowiak, K.W., Grande, I.: Bipolar Disorders. *Nature Reviews Disease Primers*. 4, 1–16 (2018).
3. Hirschfeld, R.M., Bowden, C.L., Gitlin, M.J., Keck, P.E., Suppes, T., Thase, M.E., Wagner, K.D., Perlis, R.H.: Practice Guideline for the Treatment of Patients with Bipolar Disorder (Revision). *American Journal of Psychiatry*. 159, 1–50 (2002).
4. Cohan, A., Desmet, B., Yates, A., Soldaini, L., MacAvaney, S., Goharian, N.: SMHD: A Large-Scale Resource for Exploring Online Language Usage for Multiple Mental Health Conditions. In: *Proceedings of the 27th International Conference on Computational Linguistics (COLING)*. pp. 1485–1497 (2018).
5. Coppersmith, G., Dredze, M., Harman, C., Hollingshead, K., Mitchell, M.: CLPsych 2015 Shared Task: Depression and PTSD on Twitter. In: *Proceedings of the 2nd Workshop on Computational Linguistics and Clinical Psychology*. pp. 31–39 (2015).
6. Sekulić, I., Gjurković, M., Šnajder, J.: Not Just Depressed: Bipolar Disorder Prediction on Reddit. In: *Proceedings of the 9th Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis*. pp. 72–78 (2018).
7. Xu, X., Yao, B., Dong, Y., Yu, H., Hendler, J., Dey, A.K., Wang, D.: Mental-LLM: Leveraging Large Language Models for Mental Health Prediction via Online Text Data. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*. 8, 1–32 (2024).
8. Yang, K., Zhang, T., Alhuzali, H., Ananiadou, S.: MentalLLaMa: Interpretable Mental Health Analysis on Social Media with Large Language Models. In: *Proceedings of the IEEE/ACM International Conference on Web Search and Data Mining (WSDM)*. pp. 797–806 (2024).
9. Lee, D., Jeon, H., Son, S., Park, C., An, J., Kim, S., Han, J.: Detecting Bipolar Disorder from Misdiagnosed Major Depressive Disorder with Mood-Aware Multi-Task Learning. In: *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*. pp. 4965–4977 (2024).
10. De Choudhury, M., Gamon, M., Counts, S., Horvitz, E.: Predicting Depression via Social Media. In: *Proceedings of the International AAAI Conference on Web and Social Media*. pp. 128–137 (2013).
11. Jagfeld, G., Lobban, F., Rayson, P., Jones, S.: Understanding and Improving the Role of Language in Mental Health. In: *Proceedings of the 7th Workshop on Computational Linguistics and Clinical Psychology*. pp. 165–175 (2021).
12. Harrigian, K., Aguirre, C., Dredze, M.: On the State of Social Media Data for Mental Health Research. *arXiv preprint arXiv:2011.05233*. (2021).
13. Chancellor, S., De Choudhury, M.: A Critical Review of Machine Learning Approaches for Diagnosing Mental Health Disorders from Social Media Data. *Current Opinion in Behavioral Sciences*. 36, 97–104 (2020).
14. American Psychiatric Association: Diagnostic and Statistical Manual of Mental Disorders. American Psychiatric Publishing, Arlington, VA (2013).
15. Gemini Team, Anil, R., Borgeaud, S., others: Gemini: A Family of Highly Capable Multimodal Models. *arXiv preprint arXiv:2312.11805*. (2024).
16. Cohen, J.: A Coefficient of Agreement for Nominal Scales. *Educational and Psychological Measurement*. 20, 37–46 (1960).